# Dheeraj Reddy Bhumanapalli

(+91) 7674051671 | dheerajbhumanapalli@gmail.com | github.com/dheerajreddybhumanapalli | portfolio | Hyderabad, India

## Summary

AI Software Engineer with 2+ years of experience building Generative AI solutions and production software across backend systems, data platforms, RAG, agentic workflows, and LLM inference. Experienced in taking AI concepts from client POCs to scalable implementations, including 1B+ record data platforms, NVIDIA-based LLM deployments, and reusable LangGraph architectures. Strong Python foundation with experience in FastAPI, Django, PyTorch, Hugging Face Transformers, AWS, and distributed systems.

## Technical Skills

**Programming**: Python, C++, JavaScript
**AI/ML & Generative AI**: LLMs, Multimodal LLMs, Machine Learning, Retrieval-Augmented Generation (RAG), Agentic AI, NLP, PyTorch, Hugging Face Transformers, LangChain, LangGraph, CrewAI, NVIDIA NeMo, NVIDIA NIM
**LLM Infrastructure**: vLLM, SGLang, LLM Inference, Model Serving, GPU Inference, Spec-Driven Development, GitHub Spec-Kit, BMAD-Core
**Observability & AI Security**: OpenLIT, Traceloop (OpenLLMetry), Arize Phoenix, Prometheus, Jaeger, Grafana, NVIDIA Garak, IBM AIF360
**AI-Assisted Development**: Claude Code, Cursor
**Backend**: FastAPI, Django, Flask, REST APIs
**Databases**: PostgreSQL, Elasticsearch, Redis, FAISS, ChromaDB, SQL, NoSQL
**Cloud & Distributed Systems**: AWS, Docker, EKS, ECS, S3, EC2, Celery, RabbitMQ, Asynchronous Processing, CI/CD, Git, Linux
**Web Scraping & Proxy Infrastructure**: Playwright, Selenium, BrightData, Luminati

## Work Experience

**Paltech Consulting Private Limited** — Hyderabad, India
*Software Engineer – Python Full Stack & AI* — *Jul 2024 – Present*

- Built enterprise backend and data-processing workflows using Python, Django, FastAPI, Elasticsearch, Celery, RabbitMQ, AWS, and Docker for client-facing products and large-scale ingestion systems.
- Modernized the codebase from Python 3.6 to 3.11, reducing security vulnerabilities by over 80%. Resolved a critical Elasticsearch rate-limiting issue and restored production stability.
- Delivered client POCs using NVIDIA NIM and NVIDIA NeMo, deploying models on NVIDIA GPUs and evaluating configurations for efficient LLM model serving.
- Designed a RAG POC using FAISS and ChromaDB with metadata filtering to improve retrieval relevance and response grounding. Benchmarked vLLM and SGLang under constrained GPU and RAM environments for LLM inference.
- Led a team of 5 engineers in building a Playwright-based scraping pipeline from the ground up, onboarding 150+ JavaScript-rendered international tender sources.

**Ericsson** — Bengaluru, India
*AI Engineer Intern – GAIA Team* — *Jan 2024 – Jul 2024*

- Implemented an LLM observability pipeline using Prometheus, Jaeger, and Grafana for monitoring and troubleshooting. Evaluated Arize Phoenix, Traceloop (OpenLLMetry), OpenLIT, and IBM AIF360, reducing issue identification and troubleshooting time by 70%.

## Projects

**Political Contributions Compliance Platform** — *Feb 2025 – Present*
*Python, Django, FastAPI, Elasticsearch, Celery, RabbitMQ, LangGraph, PostgreSQL, AWS*

- Built scalable ingestion pipelines for an enterprise political contributions platform, processing and indexing 1B+ records from 250–300+ U.S. government sources through APIs, Selenium, and CSV workflows.
- Modernized custom Python packages and migrated application components across 10–11 repositories, validating dependencies and application behavior through systematic testing.
- Optimized bulk processing by reducing concurrent processes from 16 to 8. Added GZIP-compressed CSV exports for 2GB+ files, reducing CPU and memory consumption and failures caused by memory exhaustion.
- Designed the HLD/LLD for the client's initial Generative AI initiative and developed a configuration-driven LangGraph pipeline with 9 reusable tools and database-managed prompts. Extended the architecture across approximately 50 sources with 70–75% reusable implementation coverage.
- Used Claude Code and Cursor to accelerate implementation, refactoring, debugging, and documentation across the platform while following existing engineering and architectural patterns.
- Automated ingestion error notifications and alerting, reducing manual monitoring effort by 70%.

**Multimodal Property Damage Detection Pipeline** — *Oct 2024 – Jan 2025*
*Python, FastAPI, React, Pixtral, Llama, LangChain, Hugging Face Transformers*

- Architected a multimodal AI pipeline with LangChain, Pixtral, and Llama for automated hail damage detection, achieving 89–90% accuracy across training and test datasets ranging from 1K–10K images.
- Implemented image classification and damage severity scoring workflows using GPU-accelerated Hugging Face Transformers with BF16 models on NVIDIA A6000 GPUs, automating first-level damage screening.

## Education

**Birla Institute of Technology and Science Pilani, Hyderabad Campus** — Hyderabad, India
*B.E. (Hons) Electrical and Electronics Engineering, CGPA: 8.35* — *Oct 2020 – Jul 2024*

## Awards & Certifications

**Claude Certified Architect – Foundations** — 2026
*Anthropic*

**Flash of Brilliance** — 2025
*Paltech Consulting Private Limited*